

Regional Growth Analysis

Task 30 – Data Analytics Track

1. Objective

The objective of this analysis is to compare regional sales growth over time and practice period-over-period comparisons using SQL and Excel. The analysis focuses on annual sales, year-over-year (YoY) growth, and sustained regional performance.

2. Dataset

A Superstore-style dataset containing 1,600 sales transactions from 2021 to 2024 was used. The dataset contains Order ID, Order Date, Region, Category, Sub-Category, Segment, Sales, Quantity, Discount, and Profit.

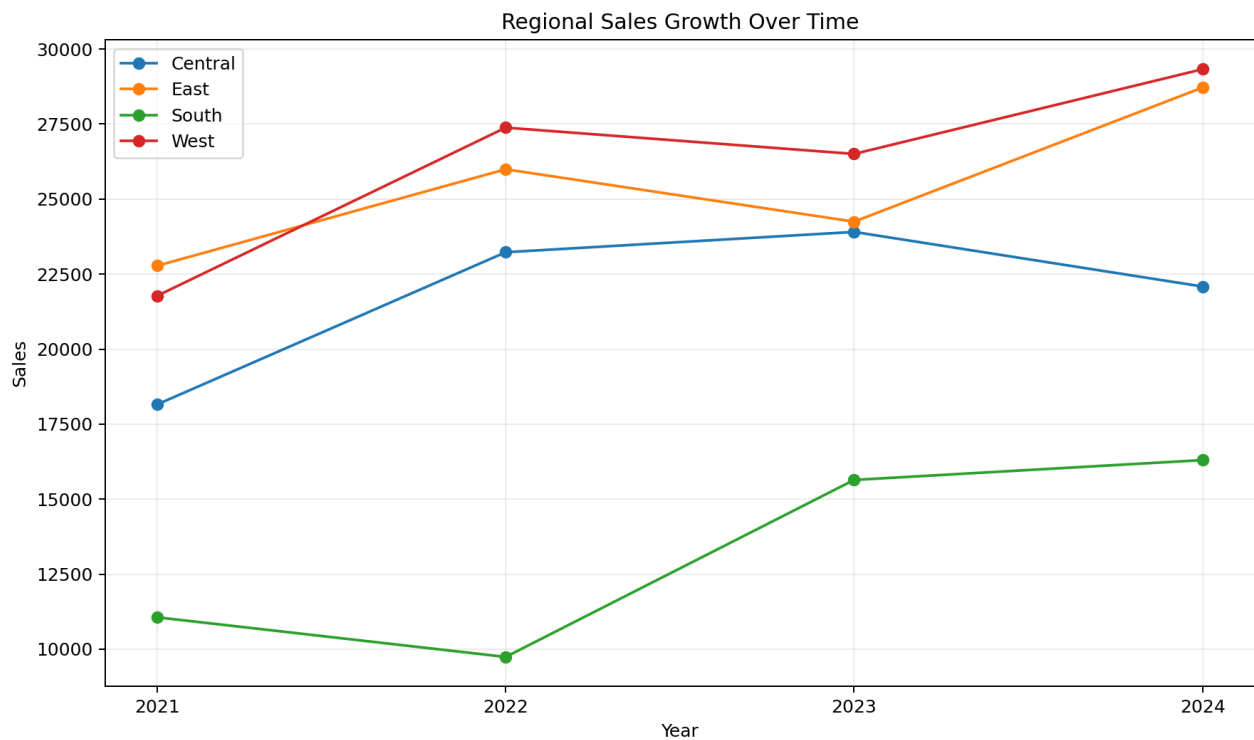
3. Methodology

- Aggregated total sales by Region and Year.
- Calculated year-over-year growth using: $((\text{Current Sales} - \text{Previous Sales}) / \text{Previous Sales}) \times 100$.
- Used SQL LAG() logic to retrieve the previous year's regional sales.
- Compared 2023–2024 growth rates across regions.
- Reviewed CAGR over the full 2021–2024 period to assess sustained growth.
- Created a regional sales trend chart and summarized five business insights.

4. Regional Growth Table

Region	Sales_2021	Sales_2022	Sales_2023	Sales_2024	Growth_2023_2024	CAGR_%
Central	18,157	23,227	23,904	22,085	-7.61%	6.75%
East	22,778	25,992	24,246	28,714	18.43%	8.03%
South	11,061	9,742	15,640	16,301	4.23%	13.80%
West	21,778	27,380	26,503	29,325	10.65%	10.43%

5. Regional Sales Growth Chart



6. Key Insights

1. West generated the highest sales in 2024, with approximately 29,325 in sales.
2. East recorded the strongest 2023–2024 growth at 18.43%.
3. Central recorded the weakest 2023–2024 growth at -7.61%, indicating a need to monitor regional momentum.
4. South had the highest CAGR from 2021 to 2024 at 13.80%, suggesting the strongest sustained expansion.
5. Growth percentage should always be evaluated together with absolute sales because a small starting base can produce a very high percentage increase.

7. SQL Analysis

The SQL deliverable uses annual aggregation and the LAG() window function to compare each region with its previous year's sales. This supports consistent period-over-period analysis without manually joining each year.

Core formula: $\text{Growth \%} = ((\text{Current Period Sales} - \text{Previous Period Sales}) / \text{Previous Period Sales}) \times 100$

8. Interview Questions

Why can growth rate mislead? Growth rate is relative to the starting value. A region with a small prior-year sales base can show a large percentage increase even when its absolute sales increase is relatively small.

What is a small-base effect? It occurs when a low starting value makes a subsequent increase look unusually large in percentage terms.

9. Business Recommendation

Management should evaluate regional performance using both absolute sales and growth percentage. High-growth regions can receive expansion attention, while regions with slowing growth should be investigated for changes in demand, product mix, pricing, discounts, or customer behavior.